

# NIRAMAI Health Analytix: Making breast cancer diagnosis easier with Google Cloud and Launchpad

Tell us your challenge. We're here to help.

[Contact us \(/contact\)](#)

ABOUT NIRAMAI HEALTH



## About NIRAMAI Health Analytix

Using TensorFlow  
learning models &  
containerization to  
Google Kubernetes  
NIRAMAI Health  
delivering a cons

Founded in mid-2016 and starting operations a few months later, India-based healthcare startup NIRAMAI Health Analytix is delivering a new machine learning- and artificial-intelligence-based breast cancer detection service. The business raised \$6 million in Series A venture capital funding in February 2019 that it plans to use for scaling operations, hiring expert staff, and obtaining the regulatory approvals needed to expand beyond India.

improving, scalable  
cancer detection  
that can complement  
traditional diagnostic  
treatment methods

Industries: Healthcare

Location: India

TensorFlow (<https://www.tensorflow.org/>),  
Google Cloud (<https://cloud.google.com/>)

Kubernetes Engine  
(<https://cloud.google.com/kubernetes-engine/>)

Kubernetes (<https://cloud.google.com/kubernetes/>)

TensorFlow (<https://www.tensorflow.org/>)  
[Contact us \(/contact\)](#)

Firebase (<https://firebase.google.com/>)

## Google Cloud Results

- Scales to support planned expansion into more locations in India and internationally
- Uses machine learning to improve the accuracy of results already at 90%
- Learns to improve user interface and how to reach users more effectively

Delivered  
accessible,  
affordable, and  
increasingly  
accurate breast  
cancer  
screening

Breast cancer is the most common cancer for women in India, with one woman in 28 likely to develop the disease, according to India Against Cancer, a group of doctors and researchers at the country's National Institute of Cancer Prevention and Research. Mortality rates from the disease in countries like India can be as high as 50 percent.

However, support for skilled entrepreneurship and new technologies are creating opportunities to reduce mortality from breast cancer through early detection and treatment – and associated personal and financial costs.

## **Artificial intelligence and machine learning key to breast cancer fight**

For India-based startup NIRAMAI Health Analytix, artificial intelligence (AI) and machine learning (ML) are key to its approach to fighting the disease.

Founded in 2016, the business is using AI and ML to enable early detection of breast cancer – and consequently help improve rates of survival from the disease. "We lose one of every two women who are detected with breast cancer, primarily because we are not able to detect it in time," says Nidhi Mathur, Chief Operating Officer and Co-founder, NIRAMAI Health Analytix. "At NIRAMAI Health Analytix, we are focused on saving those lives by detecting early."

NIRAMAI Health Analytix started by looking at ways to improve thermography, a technique that relies on

thermal imaging to detect increased warmth in the skin that overlays breast cancer lesions.

"Launchpad  
gave us an  
opportunity to  
be in an  
environment  
with 100  
mentors from all  
parts of the  
world."

—Nidhi Mathur, Chief  
Operating Officer and Co-  
founder, NIRAMAI Health  
Analytix

"We looked at the difficulties medical practitioners were experiencing interpreting these images," explains Dr. Geetha Manjunath, Co-founder, Chief Technology Officer, and Chief Executive Officer, NIRAMAI Health Analytix. "By adding AI and ML to software that analyzes thermal images, we could bring in smarter objective interpretation and data-based decision-making for this image diagnosis."

Dr. Manjunath believes NIRAMAI Health Analytix's product – which she describes as non-contact, non-invasive, and non-radiation-based, can help fill major gaps in breast cancer detection. "Mammography is a wonderful tool to detect early-stage breast cancer, but is radiation-based so multiple, frequent use isn't recommended," she says. "Furthermore, mammography is more effective for women over 45 years old – but less effective for women who have a condition called dense breasts, where more fibroglandular and supportive tissue exists than fatty tissue."

Finally, the cost of mammography machines can be a considerable burden on hospitals and medical centers in India. "Mammography machines typically cost about 1.5 crores in Indian rupees (more than US\$200,000)," says Dr. Manjunath. "Because few hospitals can afford a good machine, affordability and accessibility becomes a very important issue in a developing market like India."

## **90% accuracy**

According to Dr. Manjunath, NIRAMAI Health Analytix's product currently delivers high levels of accuracy for women with dense breasts and as well as non-dense breasts, and hence is effective for women of any age. Furthermore the scan can be performed as frequently as necessary – particularly important in cases where there is a family history of cancer.

The core of NIRAMAI Health Analytix's product is Thermalytics, a computer-aided AI and ML-based

diagnostic engine. A high-resolution thermal sensing device and cloud-hosted analytics solution combine to enable intelligent analysis of the thermal images. The business has tested its cancer screening tool on more than 7,500 women to date, while the outcomes of three clinical trials have been published in peer-reviewed journals and at top breast cancer conferences.

According to NIRAMAI Health Analytix, its trials have demonstrated:

- a 27% higher accuracy rate than mammography;
- a 70% higher positive predictive value than visual interpretation of thermography and
- effectiveness in 32% more patients who visit hospitals with pain or discomfort in their breasts.

"We learned  
about the  
Kubernetes  
container  
orchestration  
system and how  
Google  
Kubernetes  
Engine could

help us deploy  
containerized  
applications in a  
managed,  
production-ready  
environment. We  
quickly realized  
this would help  
us meet our  
scalability  
challenges."

*—Dr. Geetha Manjunath,  
Founder, Chief  
Technology Officer, and  
Chief Executive Officer,  
NIRAMAI Health Analytix*

The business uses deep learning through the

TensorFlow

(<https://opensource.google.com/projects/tensorflow>)

open source library to help determine whether the images meet its protocol requirements. From there, NIRAMAI Health Analytix uses image processing models to extract over 120 features – including vascularity structures and likely lesions – from the image, and a novel machine learning model is built to identify whether a person is likely to have breast cancer. The solution, in fact, generates a detailed report for review and certification by a doctor.

## Launchpad rockets NIRAMAI Health Analytix to success

Having developed a proof of concept to secure market acceptance, technology and clinical validation, Niramai needed support to take its business to the next level. Dr. Manjunath decided to put the organization forward for Launchpad (<https://developers.google.com/programs/launchpad/>) – a global accelerator program that helps startups scale by matching them with Google's people, networks, and technologies. "We wanted to learn how to scale our business, including our mobile and cloud infrastructure," she says.

"From a technology perspective, we wanted to accommodate traffic from the nine installations we had operating at the time, while from a business perspective, we wanted to continue to gain acceptance from doctors and clinicians who were cautious about adopting new technologies," she adds. "In addition, we wanted to understand what legal and regulatory clearances and clinical trials had to be completed to expand into markets such as the United States, Asia, and Europe."

"Launchpad gave us an opportunity to be in an environment with 100 mentors from all parts of the world," says Mathur. "We were exposed to mentors from different industries and with highly specific skills."

These mentors provided the technical and business assistance NIRAMAI Health Analytix needed to realize its ambitions. One mentor helped the



business modify the interface to its product to be more intuitive and easy to use for the clinicians and nurses who would be its primary users.

Dr. Manjunath recalled a discussion with a highly accomplished entrepreneur who changed the business' thinking about its customers. "This mentor asked us about our business models and we explained about selling our end-to-end managed solution to hospitals and diagnostic centers," says Dr. Manjunath. "He helped us realize that hospitals could be considered as simple distribution channels and there may be many more places where we could reach our end users, who were the actual beneficiaries.

"This paved the way for us to add new channels, so we established a NIRAMAI Health Analytix presence in a shopping mall. While it's still an experiment, we are enabling women who may hesitate to go to a cancer center for screening when they are well, to just visit the screening center in a mall!"

"With support  
from Google, we  
want to make  
our product fully  
accessible and  
scalable within

## India and beyond."

—Dr. Geetha Manjunath,  
Founder, Chief  
Technology Officer, and  
Chief Executive Officer,  
NIRAMAI Health Analytix

### Google Kubernetes Engine key

Finally, Launchpad mentors advised NIRAMAI Health Analytix to break its original monolithic web services architecture into smaller pieces that could seamlessly scale in parallel. "We learned about the Kubernetes (<https://cloud.google.com/kubernetes>) container orchestration system and how Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine/>) could help us deploy containerized applications in a managed, production-ready environment," says Dr. Manjunath. "We quickly realized this would help us meet our scalability challenges."

The business is running its applications and nine ML models in containers through Google Kubernetes Engine. "We've already seen improvements with Google Kubernetes Engine because we're now growing in ways that are scalable by design," says Dr. Manjunath. The business is also using the Firebase (<https://firebase.google.com/>) mobile application platform to run an application that

enables doctors to view and certify images on mobile devices.

NIRAMAI Health Analytix is retraining its ML models in a few months to account for any false positives and false negatives, typically coming from patients that present to tertiary cancer centers with difficult cases. The models are stabilizing over time.

## **Scaling seamlessly**

With its [Google Cloud Platform](https://cloud.google.com/)

(<https://cloud.google.com/>) infrastructure scaling seamlessly, the business has grown to 25 installations and over the next year, plans to expand across India. It has created two products to maximize reach – a real-time triaging report for women in rural screening camps and a detailed report for women taking the test in diagnostic centers and hospitals in urban areas. Furthermore, the business is being recognized for its achievements – Dr. Manjunath was a recent winner at the Women in AI Leadership Awards, and Niramai also won the Gold Prize in the Hack Osaka International Competition held in Japan in 2019. In February, the business raised US\$6 million in Series A venture capital funding.

"With support from Google, we want to make our product fully accessible and scalable within India and beyond," says Dr. Manjunath.

m o c e g a o o b d u o c business (

m o c e g a o o b d u o c business (